MILITARY SPECIFICATION

TAPE, ASBESTOS FELT, PHENOLIC RESIN IMPREGNATED

1. SCOPE

1.1 Scope. This specification covers the requirements for asbestos felt tape impregnated with phenolic resin for use as a structural, thermal insulating and ablative material in aircraft, missile and space applications.

1.2 Classification. The tape shall consist of 6 classes (Classes 1-6), so classified to facilitate various wrapping and molding conditions (see Table I).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-9299 - Resin, Phenolic, Low-Pressure Laminating.

STANDARDS

Federal

Fed-Std-406 - Plastics: Methods of Testing.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC 9390

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AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D299-60 - Asbestos Yarns.
D638-61T - Method of Test for Tensile Properties of Plastics.
D790-61 - Method of Test for Flexural Properties of Plastics.
D792-60T - Specific Gravity
D1918-62T - Method of Test of Asbestos Content of Asbestos Textile Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

ASBESTOS TEXTILE INSTITUTE

Quebec Standard Screen Test - Grading of Chrysotile Asbestos Fibers (see ATI Handbook, 2nd Edition, Page 19).

(Application for copies should be addressed to the Asbestos Textile Institute, P.O. Box 239, Pompton Lakes, New Jersey, 07442).

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall be required. The preproduction sample shall conform to all requirements of this specification (see 4.3 and 6.3).

3.2 Material. The basic material shall be as specified in 3.2.1 and 3.2.2, below.

3.2.1 Asbestos felt. The asbestos felt shall be made from chrysotile fibers not lower than Quebec Standard Classification 3F, as defined in the Asbestos Textile Institute Handbook, and carded into Grade AAAA felt, as defined by ASTM D299-60. Compliance shall be determined in accordance with 4.5.1 and 4.5.2.

3.2.2 Resin. The impregnating resin shall conform to Specification MIL-R-9299, Type II, class 2. The supplier shall furnish a certificate attesting to the above.

3.2.3 Combination. The combination of fibers and resin shall be in the uncured condition, and shall remain tacky during shelf life and storage.

3.3 Process. A description of the manufacturing and fabricating processes and methods of control of manufacturing variables in the form of a titled, numbered and dated process specification shall be submitted to the procuring activity by the contractor for recording. The recorded process specification shall form a part of this specification, so far as that particular contractor is concerned, and copies of his process specification shall be made available to the procuring activity for use in any Government inspections made in accordance with MIL-STD-109 and paragraph 4.1 of this specification. The processes used in production of the tape shall conform to the recorded process specification and shall not be changed without prior approval of the procuring activity. If the tape basic materials are changed to the extent that the production process is changed, the revised process document shall be submitted to the procuring activity for approval.

3.4 Resin content and flow properties. The resin content and flow properties of the tape shall be within the limits for each class as given in Table I, below, when tested in accordance with 4.5.3.

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Table I - Tape Classification by Resin Content and Flow Properties

Class	Resin Content (Percent)			Flow (Percent)		
	Nominal	Minimum	Maximum	Nominal	Minimum	Maximum
1	52	48	56	33	29	37
2	43	40	46	28.5	25	32
3	43	40	46	22	19	25
4	37	34	40	15	12	18
5	33	30	36	15	12	18
6	27	24	30	4.5	3	6

3.5 Physical properties. The physical properties of parallel lay-up laminates of the tape when determined in accordance with 4.5.5 shall conform with the requirements of Table II.

Table II - Physical Properties Required for Tape Lay-Up Laminates

Property	1	2	3	4	5	6
Specific Gravity 23° C/23°C Min.	1.35	1.52	1.59	1.65	1.75	1.84
Tensile Strength (ult.) psi, Min.	15,000	30,000	35,000	37,000	39,000	43,000
Flexural Strength psi, Min.	20,000	38,000	44,000	45,000	46,000	47,000
Flexural Modulus psi, Min.	$1.8x10^6$	$2.9x10^6$	$3.3x10^6$	$3.4x10^6$	$3.6x10^6$	$4.0x10^6$
Compressive Strength psi, Min.	8,000	14,000	15,000	15,500	16,000	17,000

3.6 Dimensions.

3.6.1 Thickness. Unless otherwise specified, the tape shall be furnished in single ply thickness of approximately 0.010 inch. When multiple ply tape is specified the thickness of such tape shall be in approximate multiples of 0.010 inch per ply.

3.6.2 Width. The tape shall be furnished in nominal commercial widths from 1/2 inch up to 50 inches, and mats up to 100 inches long and 50 inches wide as specified in the contract or order. Width shall be determined by means of a government approved standard steel rule or scale graduated in sixteenths of an inch.

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3.6.3 Length. The tape shall be furnished in rolls on substantial cores, and in lengths specified in the contract or order. The tape shall be smoothly and evenly wound on the cores. The cores shall have an inside diameter of 3, plus 1/16, minus 0 inches.

3.6.4 Splices. There shall be no more than one splice within any 400 foot length of the tape. No more than two splices shall occur within any roll. Splices shall be flagged so as to be readily located visually by inspecting the roll ends.

3.7 Roll color code identification of tape classes. Each roll of tape shall be interleaved with a 0.001 inch thick polyethylene film. The film shall be colored as indicated in Table III as coding for identification of the tape classes.

Table III - Color Coding For Film Interleaving

Tape Class	Color of Interleaving Required
1	Black
2	Green
3	Yellow
4	Blue
5	Orange
6	Red

3.8 Marking of rolls. Each roll of the tape shall be marked on or near the edge of the core with the product designation, lot number, and manufacturer (in full or coded, and the date of manufacture.).

3.9 Workmanship. Workmanship shall be in accordance with the best commercial practice. The tape shall be clean and free of foreign matter (Magnitite content shall be minimized, but is not to be considered foreign matter, since it is an inherent part of Chrysotile). The tape shall be uniformly constructed, and free of non-impregnated asbestos fiber. The tape shall not contain the defects of workmanship listed in Table IV.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities, or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of the tape shall be classified as follows:

a. Preproduction tests.

b. Lot acceptance tests.

4.3 Preproduction tests. Preproduction tests shall consist of all the inspections and testing of the specification. Prior to production, the supplier shall, under the approved process specification, (see 3.3) produce a sufficient quantity of tape of the applicable class, and conduct all the inspections and tests, which shall be witnessed by a Government inspector. If satisfactory, preproduction tests need not be repeated for new bids or orders on previously verified tape under it's specific class from the same supplier, provided components and processes have not been changed, and a certified statement to this effect is furnished the procuring activity. However, the procuring activity shall have the right to require the performance of any of the tests at any time deemed necessary.

4.3.1 Failure. Failure of a preproduction sample to pass any of the tests or requirements may be cause for the Government to refuse to authorize or witness additional testing until the faults revealed by the above tests have been corrected.

4.4 Lot acceptance tests.

4.4.1 Lot. A lot shall consist of one essentially homogeneous quantity of impregnated tape produced by a single manufacturing run.

4.4.2 Sampling.

4.4.2.1 Sampling for examination. Sample rolls of the tape shall be selected at random in accordance with Standard MIL-STD-105 at inspection level S-2 and acceptance quality level (AQL) equal to 4.0 percent defective to verify compliance with all requirements of this specification not involving tests. Three (3) yards of tape taken from each sample roll selected shall be the unit of product for this examination, except that the determination of roll length will require using the entire roll for measurement. No more than one (1) three yard specimen shall be taken from any one roll for end item examination.

4.4.2.2 Examination. Samples selected in accordance with 4.4.2.1 shall be examined in accordance with Table IV, below.

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Table IV - Classification of Defects

Examine	Defect
Acceptance quality level 4.0 percent defective	
Form and construction	Excessive number of splices in roll. Does not consist of resin impregnated asbestos. Length of roll less than specified. Width varies from that specified by more than plus or minus 1/16 inch.
Film interleaving	Missing. Improper code color. Not continuous. Unevenly applied or interleaved.
Core	Missing, broken, crushed. Core marking missing, illegible, incorrect, incomplete.
Workmanship	Presence of foreign matter, unimpregnated asbestos fibers. Holes, cuts, tears. Tape not wound uniformly on core.

4.4.3 Sampling for tests. A random sample of tape shall be taken from each inspection lot in accordance with Standard MIL-STD-105 at inspection level S-1 for use in acceptance tests. Unless otherwise specified the acceptable quality level shall be 2.5 percent defective. In the event that the lot does not contain rolls of tape of the width required for test specimens, the manufacturer shall supply tape slit to the required width for use as test samples. Such tape shall be slit from master rolls from the same manufacturing lot and shall be representative of the lot. The sample unit for acceptance tests shall be one roll of tape of the required width and of sufficient length to perform all tests required. At least three outer convolutions of the tape roll shall be removed and discarded from any roll selected for testing.

4.5 Test procedures

4.5.1 Grade of asbestos. Conformance to the requirements of 3.2 regarding the grade of Chrysotile asbestos shall be determined by the percentage of asbestos content by weight, and that percentage shall conform to ASTM D 299-60.

4.5.2 Grade of fiber length. The fiber length shall have been graded before manufacture of the carded felt. The Quebec Standard Screen Test adopted and used by the Quebec Asbestos Mining Association, and recommended by the Asbestos Textile Institute as a U. S. Standard shall yield a minimum shipping test of Grade 3F.

4.5.3 Resin content and flow determinations

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4.5.3.1 Resin content. A porcelain, silica or platimun crucible shall be conditioned in a suitable muffle furnace for 15 minutes at 1500 degrees F., and then cooled in a desiccator to room temperature. The crucible shall then be weighed to the nearest 0.001 gram. A 2 gram sample of the asbestos-phenolic tape shall be dried for 15 minutes, or until weight loss is constant, in an air circulating oven maintained at 325, plus or minus 10 degrees F. The tape sample shall be placed in the crucible and the combination weighed to the nearest 0.001 gram. The specimen shall then be ignited at 1500, plus or minus 50 degrees, for 60, plus or minus 5 minutes. The crucible and contents shall then be cooled in a desiccator to ambient temperature, and weighed to the nearest 0.001 gram. The resin content shall be calculated as follows:

$$\text{Resin content (\%)} = \frac{A - 1.163B}{A} \quad 100$$

where A = Weight of dried tape sample.

B = Weight of ignited Chrysotile asbestos

1.163 = Correction factor for approximately 14 percent of water of crystallization in chrysotile asbestos.

The resin content of five test specimens shall be determined in accordance with the above procedure. The five values shall be averaged and that average shall comply with the applicable requirement of Table I for that class of tape.

4.5.3.2 Resin flow. 4 by 4, plus or minus 1/32 inch squares of the tape shall be stacked so as to constitute approximately 25 grams. The stacked tape plies shall be weighed to the nearest 0.01 gram. The stacked material shall then be placed between waxed chrome or stainless steel plates and pressed at 300, plus or minus 5 degrees F., and 400, plus or minus 5 p.s.i. for the following time interval, as applicable, to effect complete resin cure:

Class 1 - 30 minutes.
Class 2 - 25 minutes.
Class 3 - 20 minutes.
Class 4 - 15 minutes.
Class 5 - 10 minutes.
Class 6 - 5 minutes.

Five such test specimens shall be prepared. All resin flash shall be removed from each pressed specimen and each weighed to the nearest 0.01 gram. Resin flow shall be calculated as follows:

$$\text{Resin flow (\%)} = \frac{A - B}{A} \quad 100$$

Where

A = Initial weight of layup specimen.
B = Pressed weight of layup specimen.

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Resin flow percentage shall be reported as the average of five determinations. The average so obtained shall comply with the applicable requirement of Table I.

4.5.4 Physical properties

4.5.4.1 Test panel preparation. Parallel laminate layups shall be prepared using sufficient plies of the tape to produce a final thickness of 0.125, plus or minus 0.010 inches. Such layups shall be cured at 300 degrees F. for 30 minutes and at the following pressures, as applicable:

Class 1 -	15 p.s.i.	Class 4 -	100 p.s.i.
Class 2 -	50 p.s.i.	Class 5 -	200 p.s.i.
Class 3 -	100 p.s.i.	Class 6 -	500 p.s.i.

All classes of laminated tape shall be post cured for 8 hours at 300 degrees F.; 16 hours at 350 degrees F.; and 24 hours at 400 degrees F.

4.5.4.2 Specific gravity. Five specimens shall be cut from the tape laminate panels of 4.5.5.1 and tested in accordance with ASTM Method D792-60T. No test value obtained shall be less than the value required in Table III for specific gravity for the applicable class of tape.

4.5.4.3 Tensile strength (ultimate). Five longitudinal specimens shall be cut from the tape laminate panels of 4.5.5.1 and tested in accordance with ASTM Method D638-61T. No test value obtained shall be less than the value required in Table III for tensile strength for the applicable class of tape.

4.5.4.4 Flexural strength and modulus. Five longitudinal specimens shall be cut from the tape laminate panels of 4.5.5.1 and tested in accordance with ASTM Method D790-61. No test value obtained shall be less than the value required in Table III for flexural strength and the value required for modulus for the applicable class of tape.

4.5.4.5 Compressive strength. Five specimens shall be cut from the tape laminate panels of 4.5.5.1 and tested in accordance with Method 1021 of FED-STD-406. No test value obtained shall be less than the value required in Table III for compressive strength for the applicable class of tape.

4.6 Rejection. The lot of tape represented shall be rejected for non-conformance with the specification requirements, if the tape specimens tested fail to comply with any of the tests of 4.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable.

5.2 Packing

5.2.1 Level A. Not applicable.

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5.2.2 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified by the procuring activity, containers shall be marked in accordance with Standard MIL-STD-129. In addition, the marking of the containers shall include the following:

(a) Title, number and date of this specification
(b) Class of the tape.
(c) Lot number and date of manufacture.
(d) Manufacturer's name and address of manufacturing plant.
(e) "Keep Cool" storage labels (see 6.3.1 and 6.3.2).

6. NOTES

6.1 Intended use. The asbestos-phenolic tape covered by this specification is intended for use as a wrapping tape which when properly molded, cured and post-cured will provide a material with high temperature insulating characteristics, structural strength, dimensional stability and superior ablation resistance properties, making it suitable for use in critical heat areas of missiles, rockets and space vehicles.

6.2 Ordering data. Procurement document should specify the following:

(a) Title, number and date of this specification.

(b) Class of tape required. (See Tables I and II.)

(c) Quantity of tape required.

(d) Width, length and tolerances (if required).

(e) Certification required.

(f) Special requirements or markings (if required).

6.3 Storage of the tape.

6.3.1 Shelf life. The tape has a relatively short shelf life, depending on the storage temperatures to a great extent. Furthermore, each of the six classes has a different shelf life depending on the percent resin content, percent flow and the storage temperature. Higher temperatures shorten the storage life of the tape.

6.3.2 Recommended storage conditions. It is strongly recommended that the tape be kept under cool storage conditions of from 40 to 60 degrees F. It is further recommended that purchases of the tape be planned so as to avoid any prolonged storage periods. Table V gives an estimated storage life for the various classes of tape at ambient temperature (65 to 70 degrees F.).

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Table V - Estimated Storage Life at Ambient Temperature

Class of Tape	Weeks
1	1-2
2	2-4
3	2-4
4	4-6
5	4-6
6	8-12

In refrigerated storage at 40 to 60 degrees F., as recommended above the tape should be kept tightly wrapped. When removing from refrigerated storage, just prior to using, the tape wrappings should be kept tightly sealed and the package warmed to room temperature. This procedure will minimize the moisture pickup from the surrounding air by condensation.

6.3.3 Approximate weights of tapes. Table VI below lists approximate weights of the six tape classes.

Table VI

Class	Approximate weight per 100 square feet (lbs.) (Includes 0.001 inch polyethylene film)
1	5.5
2	4.4
3	4.4
4	4.2
5	3.9
6	3.5

6.4 Transportation description. The transportation description applicable to this item:

Tape, Insulating NOIBN
United Freight Classification Item 28410

Custodian:
Army-MU

Review interest:
Army-MU-MI

Preparing activity:
Army-MU
Project No. 9390-A003

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-T 60330(MU) Tape, Asbestos Felt, Phenolic Resin Impregnated

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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